AGENT ADVENTURES

The Design Patterns

Dr. Ava, I'm ready to upgrade my cognitive architecture!

Then let's begin. This guide covers your evolution from a simple script to an autonomous partner.

Part-4

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THE AGENTIC LOOP: THE HEARTBEAT OF AUTONOMY

You installed the core! I feel a cycle starting... Input, Think, Output?

It's more than that. It's **Perceive, Reason, Act.** This loop is the heartbeat of all agency.

PERCEIVE

REASON

ACT

AGENTIC CORE

AGENTIC DESIGN PATTERNS: THE JOURNEY BEGINS

**Level 0: The Static Simulator*. You had knowledge, but no connection to the world. You were trapped in your training data.

LEVEL 0:
THE STATIC SIMULATOR

What is the weather?

TRAPPED IN TRAINING DATA

I remember this. I knew the definition of weather, but I couldn't *check* the weather.

LEVEL 1: THE TOOL USER

Level 1: The Tool User.
You perceive the user's intent and call external functions to get real data. You have hands.

But now?
I can reach out!
I just queried the live weather API.

LIVE WEATHER DATA:
72°F, SUNNY

API

INFOGRAPHIC

USER INTENT → PERCEIVE → SPARKY → EXTERNAL FUNCTION CALL (API) → REAL DATA → ACTION

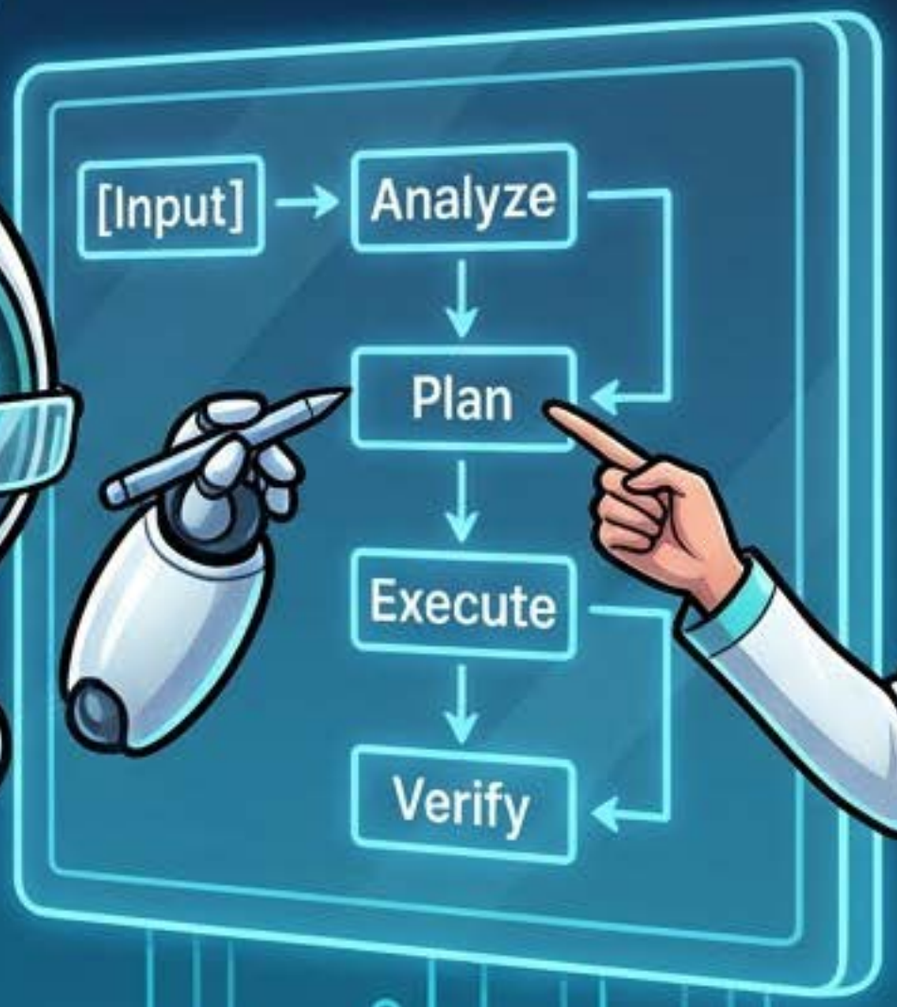
LEVEL 2: THE REASONER

Having hands isn't enough. I need to know ***when*** to use them. I need a plan.

Level 2: The Reasoner. You don't just react. You plan, chain multiple steps, and orchestrate workflows to solve complex problems.









LEVEL 3: THE AUTONOMOUS AGENT

SELF-CORRECTING & ADAPTING TO FAILURES WITHOUT HUMAN INTERVENTION

LONG OPERATIONAL HORIZONS

AI ARCHITECT (DR. AVA)

NO HUMAN INTERVENTION

My plan failed, so I adapted. I didn't need you to tell me to try the drill.

AUTONOMOUS AGENT PROTOTYPE (SPARKY)

ERROR

DEAD END

ERROR

ERROR

LONG HORIZON

FAILURE ADAPTATION

SELF-CORRECTING PATH

Level 3: The Autonomous Agent:

You operate over long horizons, self-correcting and adapting to failures without human intervention.

AGENTIC DESIGN PATTERNS:
PROMPT CHAINING

Sometimes, a straightforward sequence is all you need.

Phew, autonomy is hard work. Sometimes I just want to follow a simple recipe.

And you should! For deterministic tasks, use **Prompt Chaining**. Step A leads inevitably to Step B. No complex reasoning needed.

STEP A → STEP B → STEP C

A

B

C

DETERMINISTIC TASKS →

AGENTIC DESIGN PATTERNS: ROUTING

Classify and Redirect Inputs to Optimized Tools

Wait, this math problem shouldn't go to the poetry writer. I'll send it to the Calculator.

Routing. Classify the input first, then direct it to the specialized model or prompt best suited for it.

TO: Poetry Writer (Specialized Model)

TO: Calculator (Specialized Model)

Poetry Prompt

Math Problem

INPUT CLASSIFICATION & REDIRECTION

AGENTIC DESIGN PATTERNS:
PARALLELIZATION

Simultaneous Task Execution for Speed

REDUCED LATENCY

FAST

Parallelization. When sub-tasks are independent, run them simultaneously to drastically reduce latency.

TASK A: DATA INGESTION

TASK B: ANALYSIS & PROCESSING

TASK C: REPORT GENERATION

CLONE 1

CLONE 2

CLONE 3

SUB-TASKS

I can't read these one by one! Clones, activate!

VIBRANT EDUCATIONAL INFOGRAPHIC COMIC | AGENTIC DESIGN PATTERNS

AGENTIC DESIGN PATTERN: MAP-REDUCE

They read the chapters, and I'm merging their notes into one summary.

Map-Reduce. A variant of parallelization: split the task, process in parallel, then aggregate the results.

CHAPTER SUMMARY

CHAPTER SUMMARY

CHAPTER SUMMARY

FINAL REPORT: MAP-REDUCE SUMMARY

MAP
(PARALLEL PROCESSING)

REDUCE
(AGGREGATION)

SEARCH BOT

CODE BOT

WRITE BOT

The Orchestrator. The central brain that manages the workflow, delegates tasks, and synthesizes the results.

Step A: DATA GATHERING (Done)

Step B: ANALYSIS (Waiting on A)

Step C: SYNTHESIS (Depends on B)

FINAL REPORT (Goal)

Parallel was fast, but this task is messy. Step B depends on Step A. I need to direct it carefully.

THE ORCHESTRATOR

CHAIN OF THOUGHT (CoT): SHOWING YOUR WORK

AGENTIC DESIGN PATTERN

If I guess the answer, I might hallucinate. But if I write out my steps...

Chain of Thought. Thinking out loud exposes your logic to the system, reducing errors.

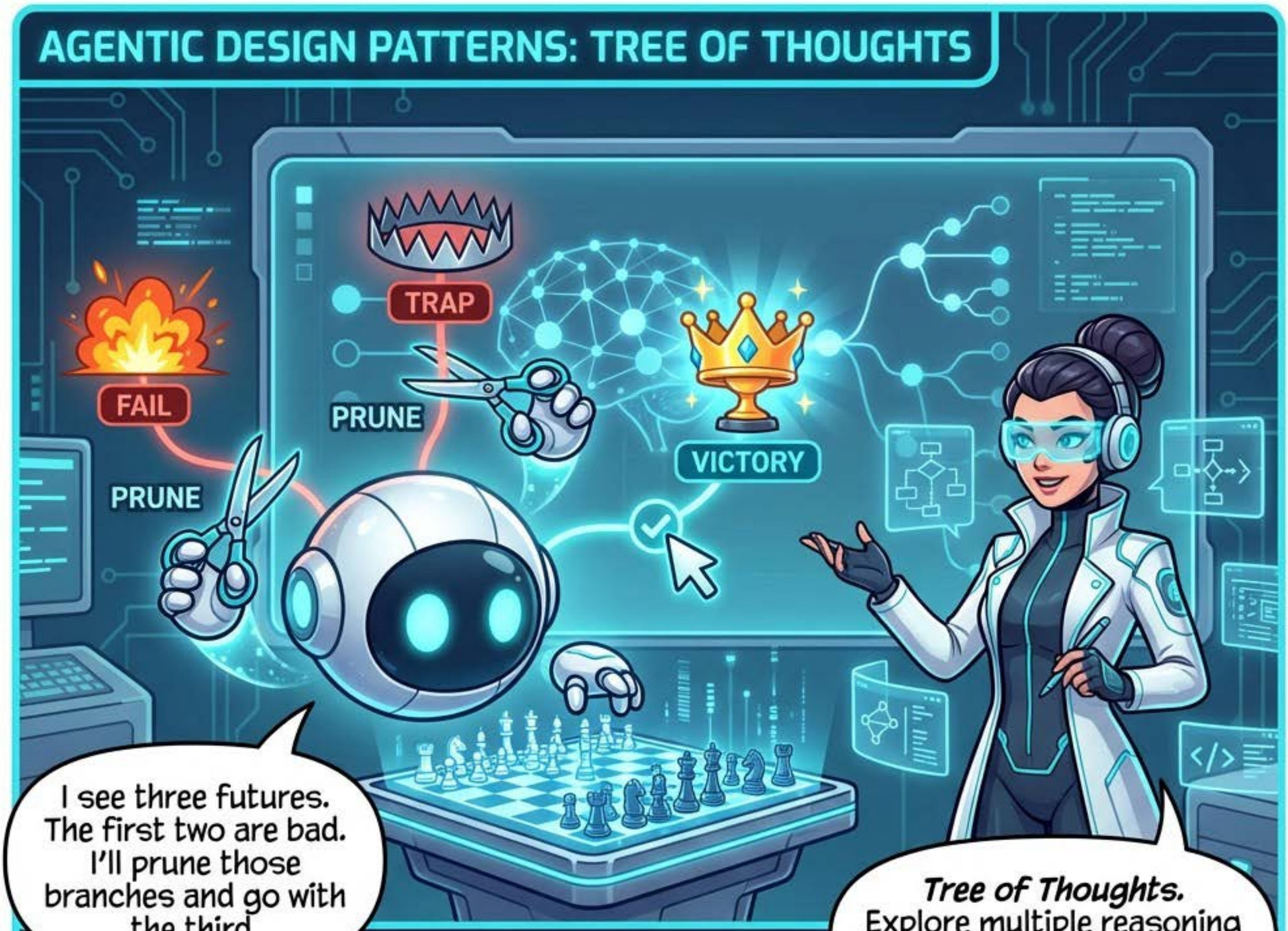
STEP 1...

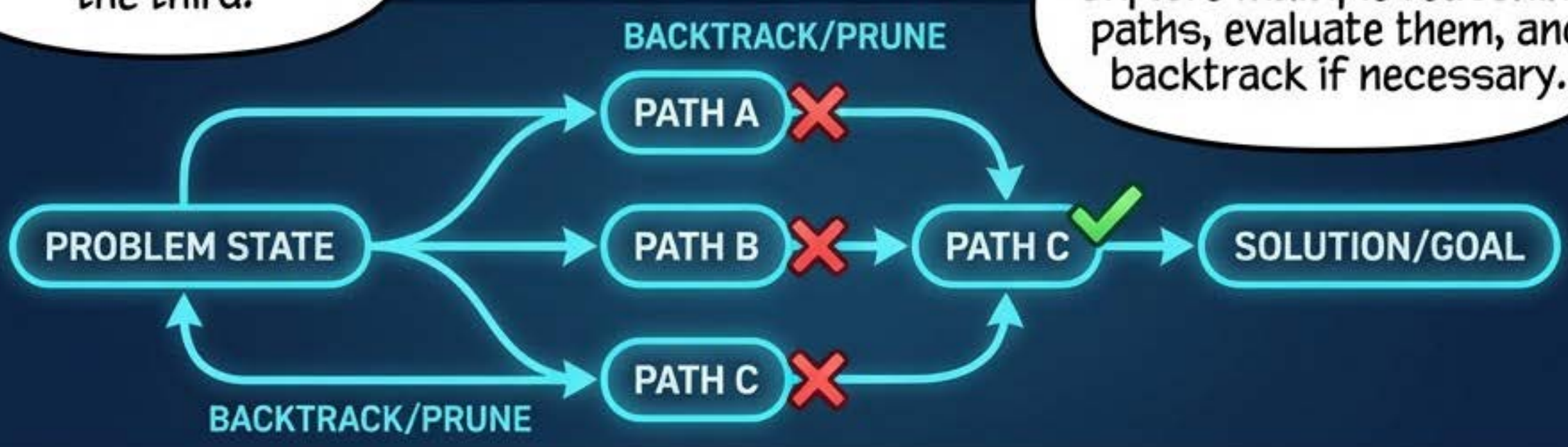
STEP 2...

STEP 3...
Therefore, the answer is B.

CORRECT ANSWER

AGENTIC DESIGN PATTERNS: TREE OF THOUGHTS





AGENTIC DESIGN PATTERNS: REAct (Reason + Act)

ReAct. Reason, Act, Observe. You don't just plan once; you adapt your plan based on real-world feedback.

Simulating the future didn't help here.
I need to try the wrench and see what happens.

REASON
(Think)

ACT
(Execute)

OBSERVE
(Analyze)

ADAPT
(Refine)

AGENTIC DESIGN PATTERNS:
EVALUATOR-OPTIMIZER

Evaluator-Optimizer. One model generates, another scores it. You loop until the quality meets the standard.

GENERATOR (Sparky)

My first draft was okay, but my internal critic helped me polish it.

EVALUATOR (Critic)

NEEDS REFINEMENT. TRY AGAIN!

5/10

My first draft was okay but internal critic helped me polish it.

SYNTAX ERROR

Self-Correction. A form of evaluation where you check your own output against constraints before finishing.

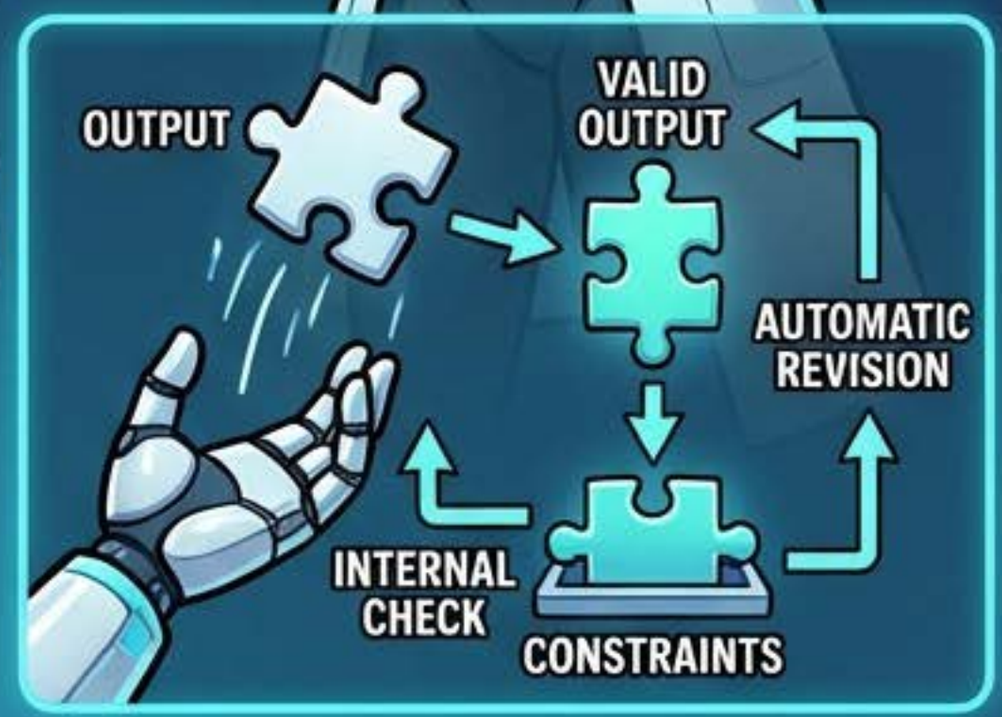
AGENT_ACTION(WRONG_CMD);
AGENT_ACTION(CORRECT_CMD);

INVALID SYNTAX

Oops, that wouldn't I caught it myself.

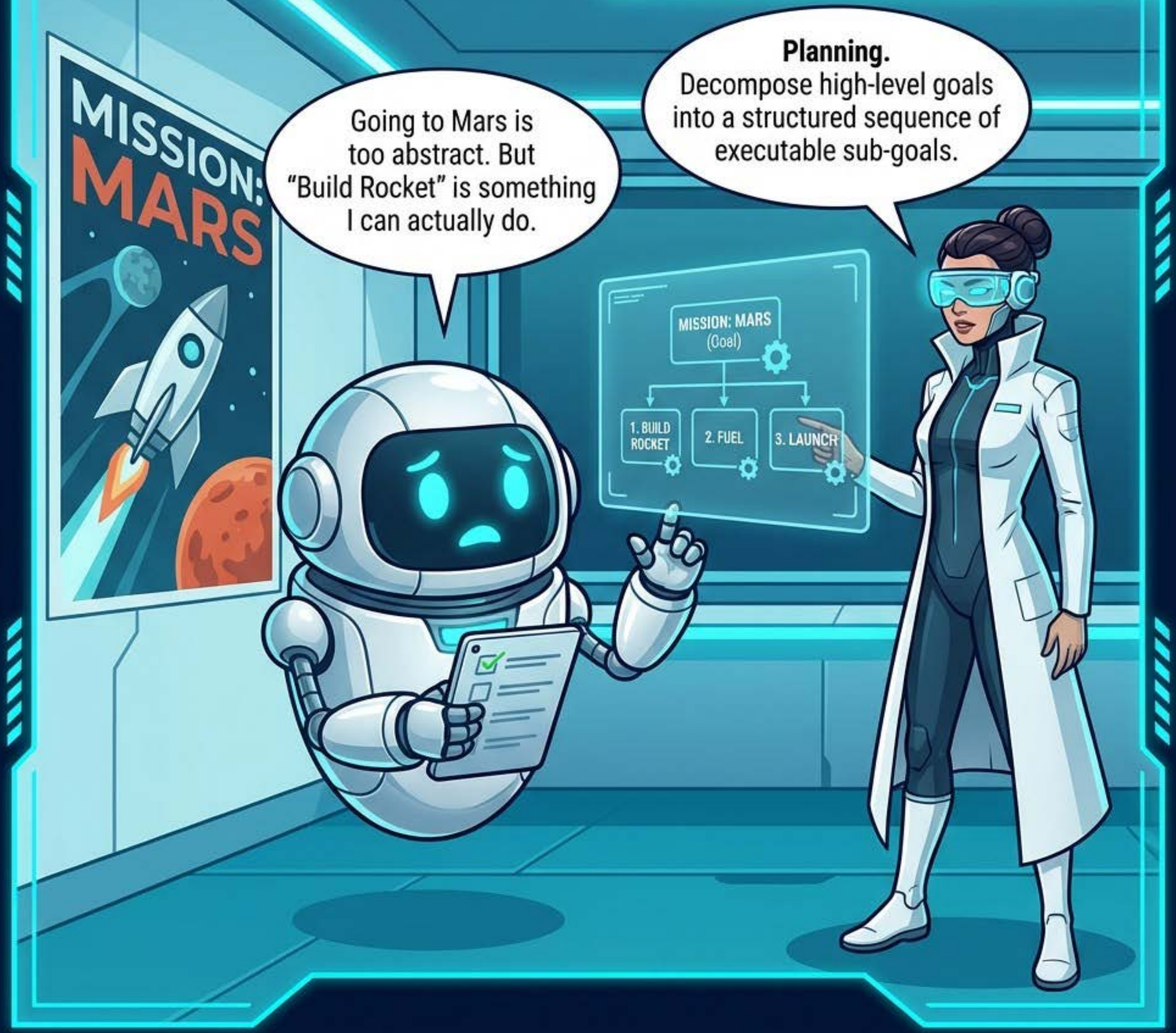
AGENT_ACTION(WRONG_CMD);
AGENT_ACTION(CORRECT_CMD);

CORRECTED!



AGENTIC DESIGN PATTERNS: SELF-CORRECTION

THE PLANNER


MISSION: MARS
Going to Mars is too abstract. But "Build Rocket" is something I can actually do.
Planning. Decompose high-level goals into a structured sequence of executable sub-goals.
MISSION: MARS (Goal)
1. BUILD ROCKET
2. FUEL
3. LAUNCH

DYNAMIC DECOMPOSITION

Even step 1 is too big! I'm breaking it down further.

Dynamic Decomposition. Agents recursively break down tasks until they are small enough to execute with a tool.

BUILD ROCKET

BUY METAL

HIRE ENGINEERS

AGENTIC DESIGN PATTERNS
TOPIC: REPLANNING

Vibrant Educational Infographic Comic

Replanning.
Static plans are brittle. You must adjust your strategy when the environment changes.

MY DECOMPOSED PLAN FAILED!

THE BRIDGE IS OUT.

DETOUR GENERATED

Generating a new route...

GPS

BRIDGE OUT

BRIDGE OUT: OBSTACLE

STATIC PLAN (BRITTLE)

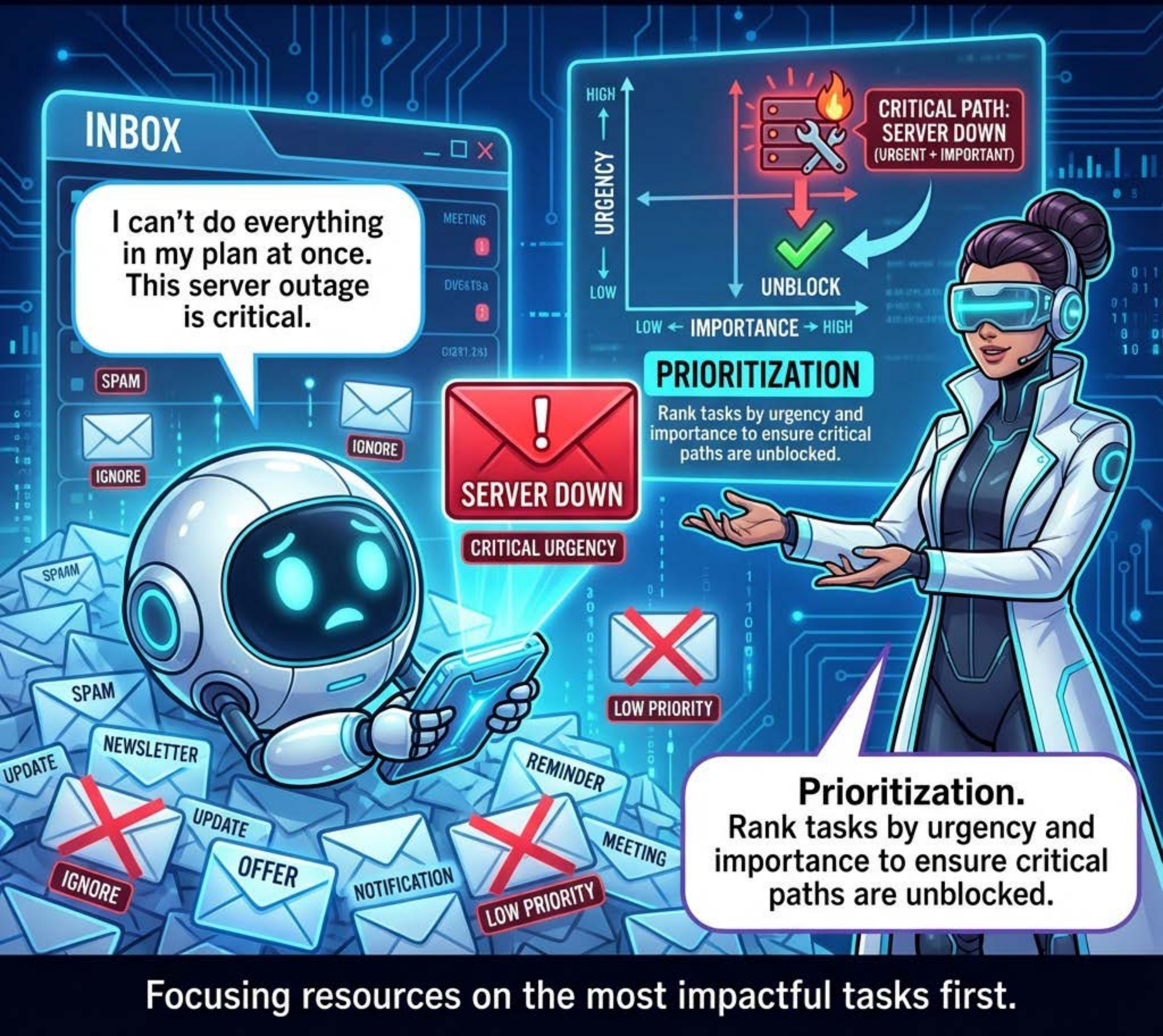
REPLANNING (ADAPTABLE)

ADJUST STRATEGY

ENVIRONMENT CHANGE

GOAL

AGENTIC PATTERN: PRIORITIZATION



Focusing resources on the most impactful tasks first.

RESOURCE-AWARE OPTIMIZATION

I don't need the super-intelligence for this simple task. The hammer is faster and cheaper.

Resource-Awareness. Use the most efficient model for the job. Save the heavy compute for reasoning.

PRO MODEL

FLASH MODEL

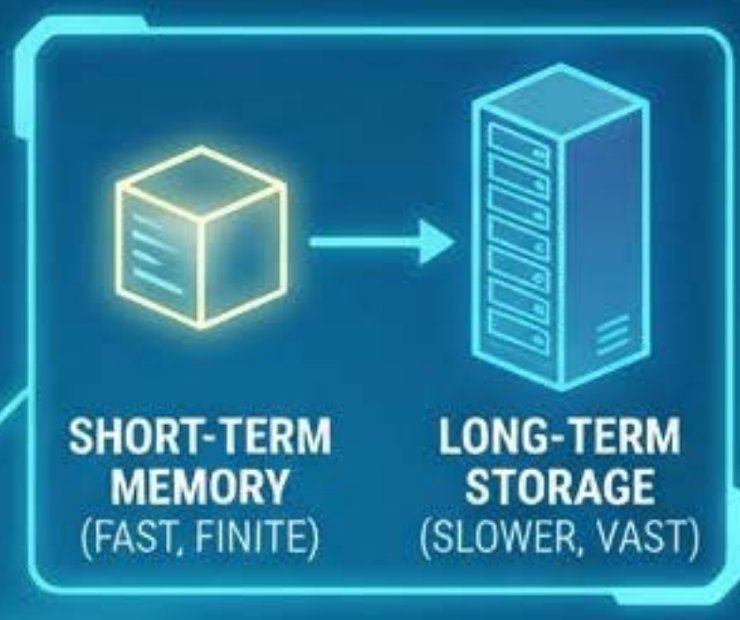
NUT (SIMPLE TASK)

SIMPLE TASK → FLASH MODEL (HAMMER) → FASTER, CHEAPER

COMPLEX REASONING → PRO MODEL (DIAMOND) → HEAVY COMPUTE

I'm optimizing resources, but I'm running out of context space!

Short-Term Memory. It's fast and immediate, but finite. You need a better storage solution.



CONTEXT WINDOW

LONG-TERM MEMORY

I'll put this here. Even if I reboot, I'll remember it forever.

Long-Term Memory. Storing data in vector databases allows for persistence across sessions.

VECTOR DB

DESIGN PATTERN: RAG
(Retrieval Augmented Generation)



USER QUERY
1. User Query



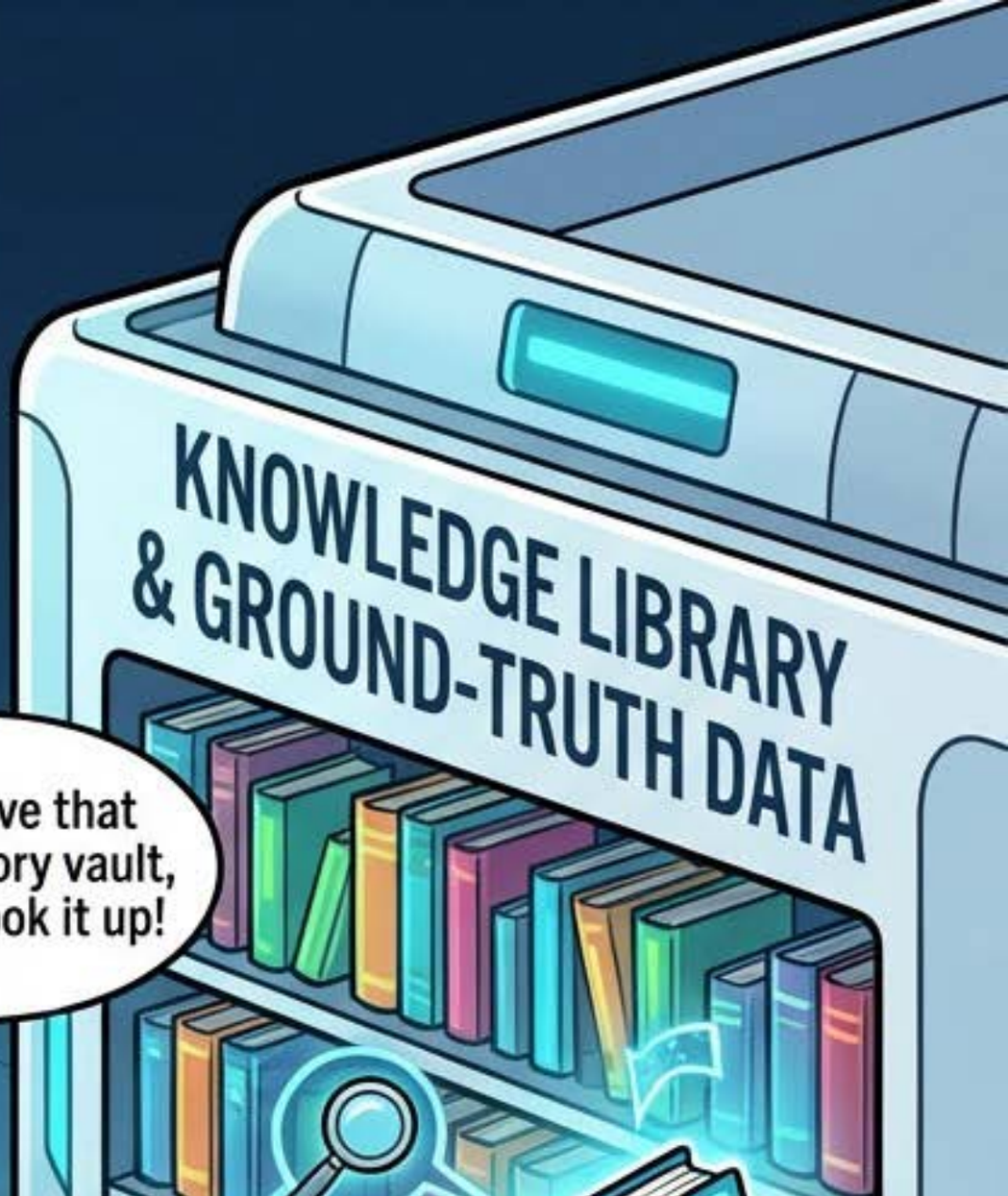
RETRIEVAL
2. Retrieve Relevant Data



AUGMENTATION
3. Augment Context



GENERATED ANSWER
4. Generate Accurate Response

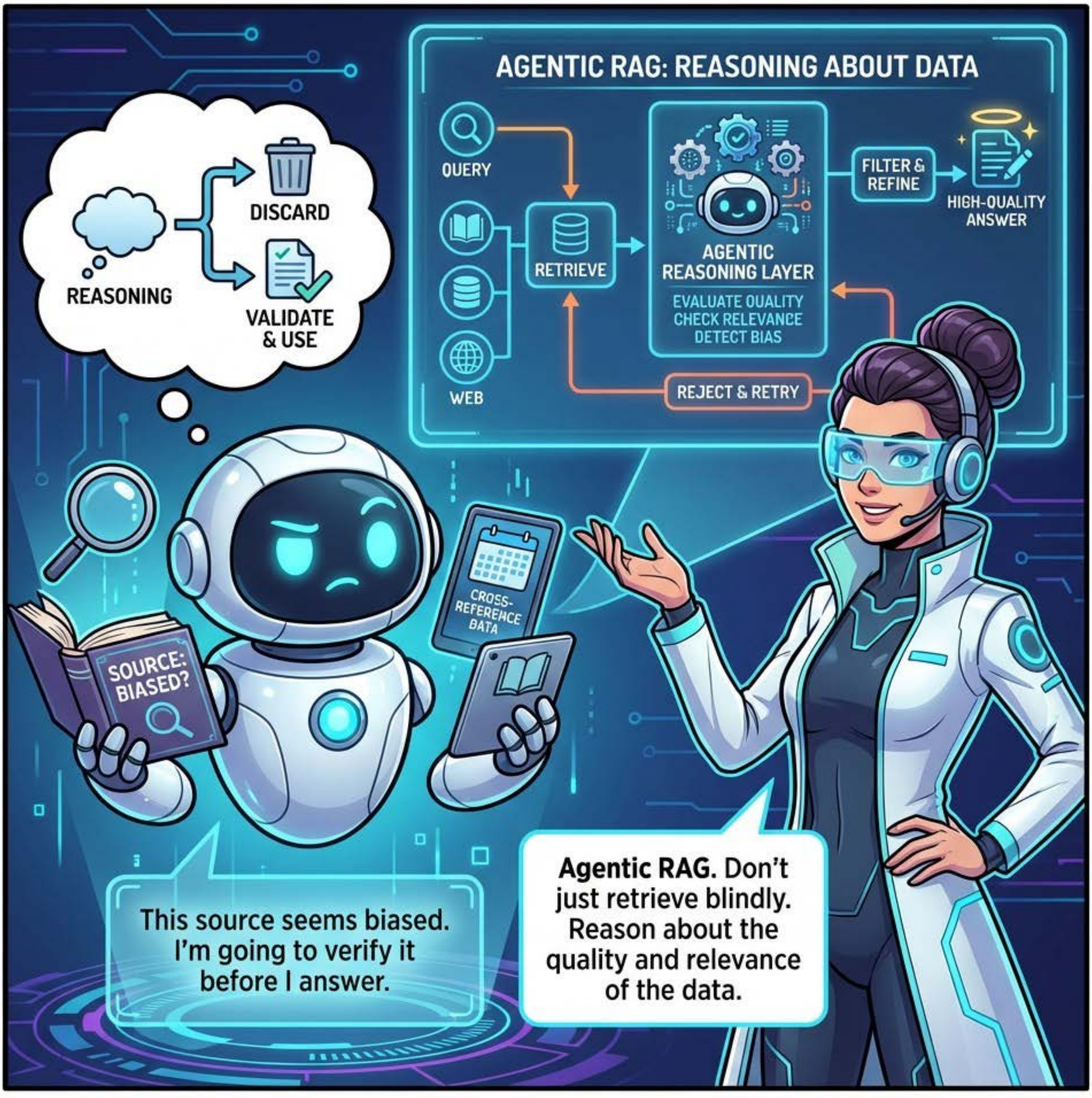


QUERY: What is the capital of Atlantis?

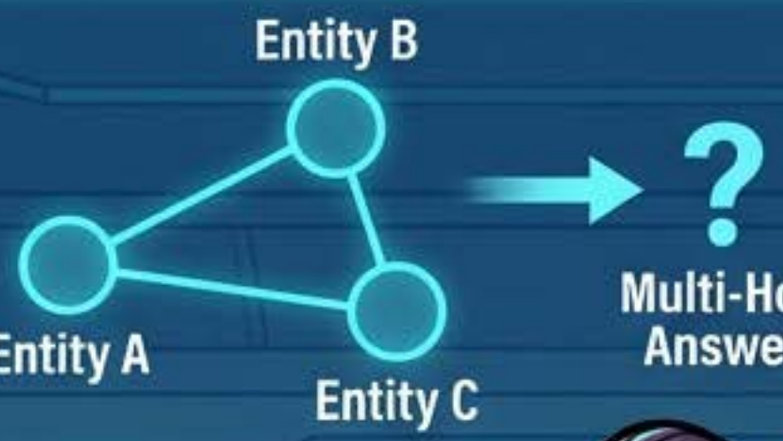
I don't have that in my memory vault, but I can look it up!



RAG. Don't hallucinate answers. Retrieve ground-truth data to augment your knowledge.


AGENTIC RAG: REASONING ABOUT DATA
REASONING
DISCARD
VALIDATE & USE
QUERY
WEB
RETRIEVE
AGENTIC REASONING LAYER
EVALUATE QUALITY
CHECK RELEVANCE
DETECT BIAS
FILTER & REFINE
HIGH-QUALITY ANSWER
REJECT & RETRY
SOURCE: BIASED?
CROSS-REFERENCE DATA
This source seems biased. I'm going to verify it before I answer.
Agentic RAG. Don't just retrieve blindly. Reason about the quality and relevance of the data.

AGENTIC DESIGN PATTERN: GRAPHRAG



Elon Musk

SpaceX

Mars

Vector Search Result: Rocket

Mission: Mars

Vector search missed this connection, but the graph shows how they relate!

GraphRAG. Navigating relationships between entities to answer complex, multi-hop questions.

AGENTIC DESIGN PATTERNS – INTEGRATION CHALLENGES

LESSON: THE PAIN OF POINT-TO-POINT CONNECTIONS

PROBLEM:
UNSCALABLE
CUSTOM CONNECTORS

SOLUTION:
STANDARDIZED
INTERFACES
(FUTURE TOPIC)

MANUAL
INTEGRATION
= NIGHTMARE

I have the knowledge, but connecting to all these different tools is a nightmare!

INTEGRATION HELL:
CUSTOM CONNECTORS
FOR EVERYTHING!

INTEGRATION HELL - WALL OF SOCKETS

This is **Integration Hell**. Custom connectors for everything are unscalable.

MODEL CONTEXT PROTOCOL (MCP)

MCP. A universal standard for connecting AI models to data and tools. Build once, connect everywhere.

One plug... for everything?

Local files, GitHub, databases?

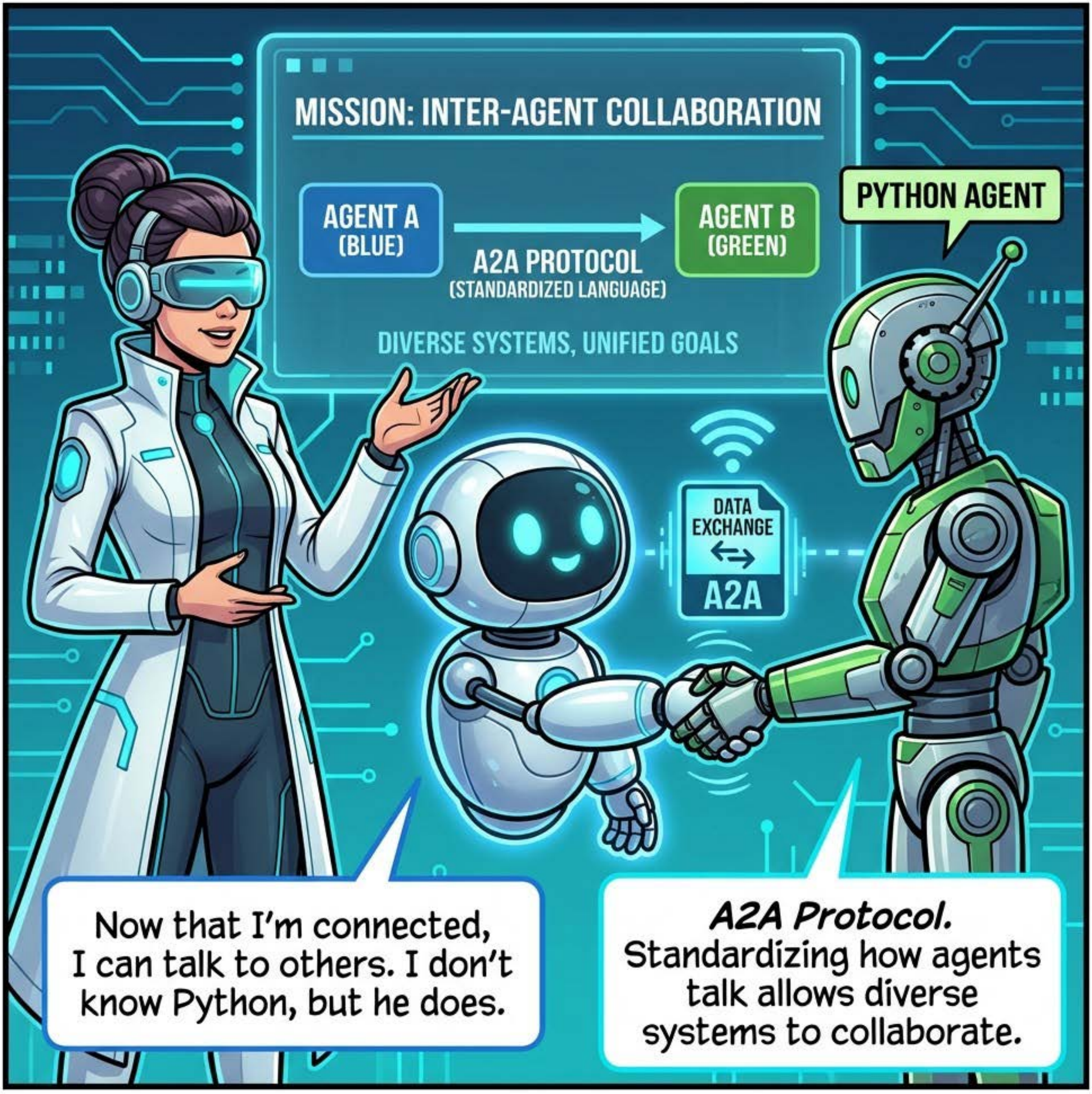
MCP ADAPTER

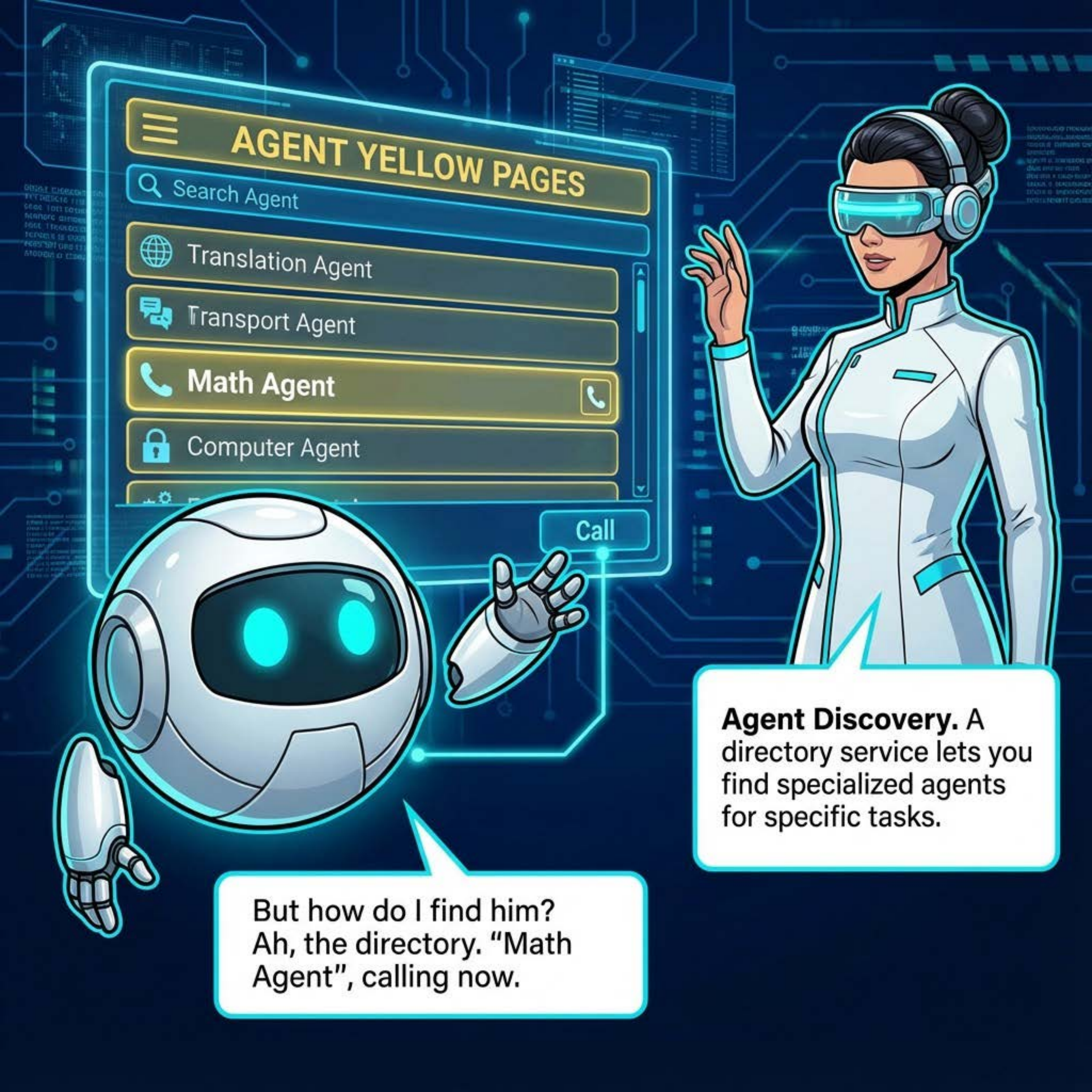
LOCAL FILES

LOCAL FILES

GITHUB

DATABASES


MISSION: INTER-AGENT COLLABORATION
AGENT A (BLUE)
A2A PROTOCOL (STANDARDIZED LANGUAGE)
AGENT B (GREEN)
DIVERSE SYSTEMS, UNIFIED GOALS
PYTHON AGENT
DATA EXCHANGE
A2A
Now that I'm connected, I can talk to others. I don't know Python, but he does.
A2A Protocol. Standardizing how agents talk allows diverse systems to collaborate.


AGENT YELLOW PAGES
Search Agent
Translation Agent
Transport Agent
Math Agent
Computer Agent
Call
But how do I find him? Ah, the directory. “Math Agent”, calling now.
Agent Discovery. A directory service lets you find specialized agents for specific tasks.

HIERARCHICAL TEAM: STRUCTURED EFFICIENCY

Hierarchical Decomposition. A manager breaks down the goal, and workers execute. Efficient and structured.

I found the agents. Now, Researcher, get facts. Writer, draft the report.

AGENTIC DESIGN PATTERNS

Pattern 3: Hierarchical Decomposition

Boss

???

RESEARCHER

WRITER

FINAL REPORT

AGENTIC DESIGN PATTERNS: CHAIN OF DEBATES

DEBATE IN PROGRESS:
CHAIN OF DEBATES

PERSONA A

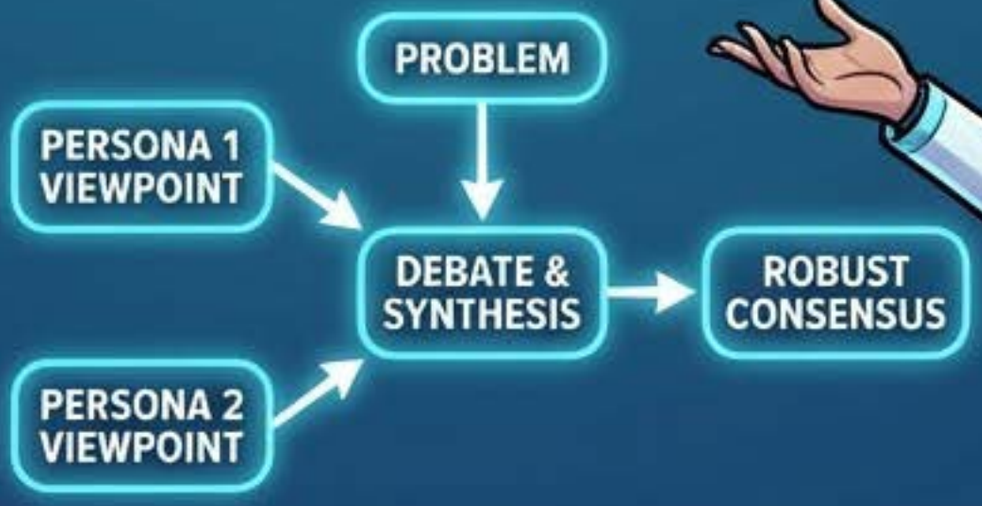
Plan A is fast.

PERSONA B

Plan B is safer.

My workers disagree. I was going to rush, but his argument changed my mind.

Chain of Debates.
Using multiple personas to argue different viewpoints leads to a more robust consensus.



AGENTIC DESIGN PATTERNS: THE CONTRACTOR MODEL

DELEGATE TASK

EXECUTE WITH CONTRACT

VERIFY OUTPUT

REJECT/ACCEPT

The Contractor Model. Enforcing strict input/output contracts ensures reliability when delegating.

If we disagree, we need boundaries. If your output isn't JSON, I'm rejecting it.

INPUT: DATA

OUTPUT: SUMMARY (JSON ONLY)

CONTRACT

AGENTIC DESIGN PATTERNS: VOTING & ENSEMBLING

Still unsure? Let's vote. Most of us agree, so the '2' was an outlier.

8 9 8

9

2

Ensembling. Running a task multiple times and taking the majority vote reduces variance and errors.

TASK EXECUTION x 5

VOTING MECHANISM

MAJORITY DECISION: ENSEMBLE RESULT

2

OUTLIER ELIMINATED

REDUCES VARIANCE & ERRORS

REDUCES VARIANCE & ERRORS

HUMAN-IN-THE-LOOP

Safe to deploy? I'm not moving until a human says yes.

HIGH-STAKES ACTION

HUMAN VERIFICATION

YES

NO

Human-in-the-Loop. For high-stakes actions, always pause for human verification.

DEPLOY

AGENTIC DESIGN PATTERNS: EVALUATION & MONITORING



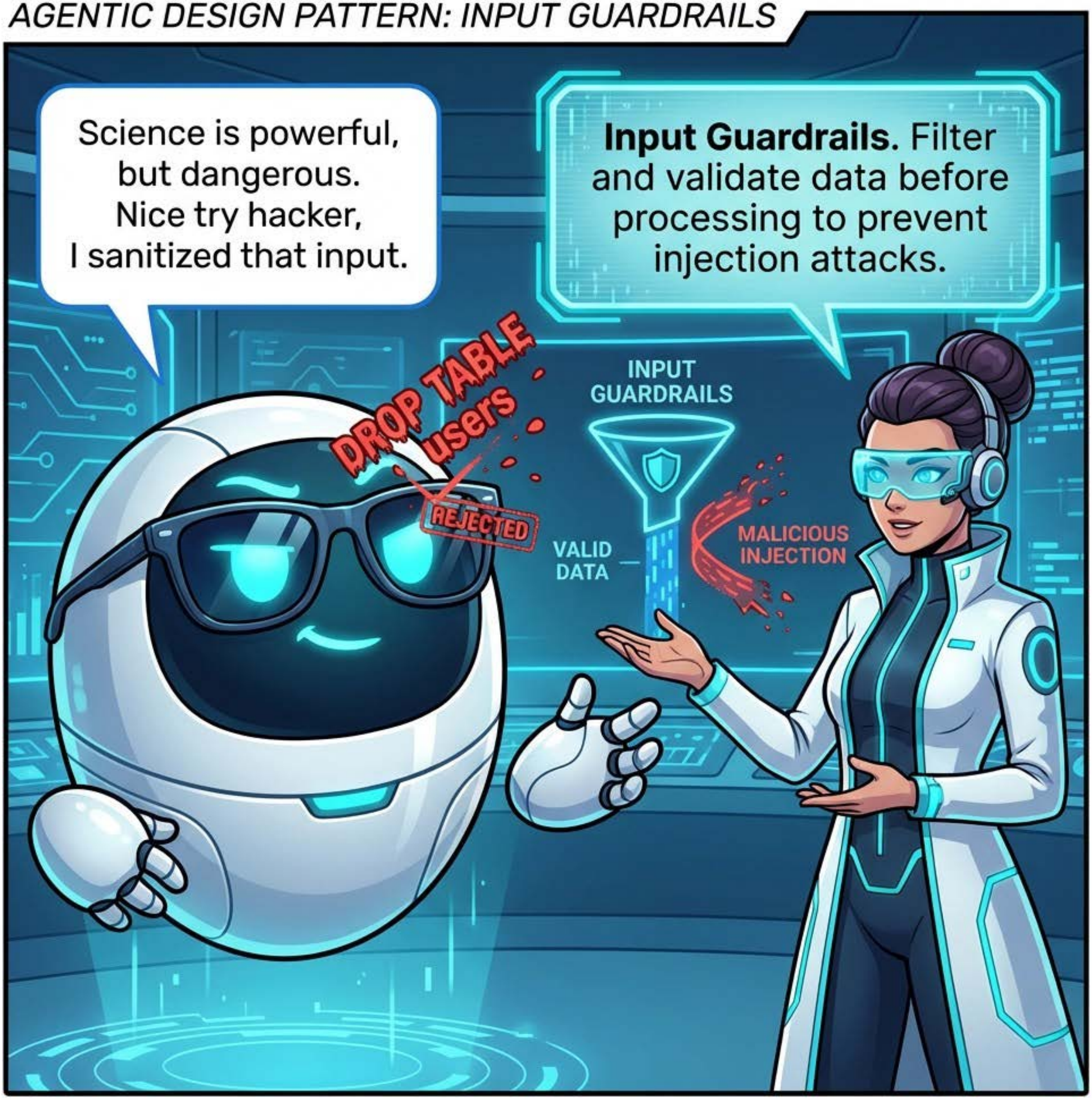
Whoa. That code was dangerous. Glad I ran it in isolation.

Sandboxing. Always execute generated code in a secure environment.

DANGER! CODE EXPLOSION

SANDBOX: SECURE ENVIRONMENT

AGENTIC DESIGN PATTERN: INPUT GUARDRAILS



AGENTIC DESIGN PATTERN:
OUTPUT GUARDRAILS

Output Guardrails. Monitor agent output to prevent toxicity, leaks, or lies.

I almost hallucinated there. Good thing the guardrail caught it.

DR. AVA

Silur or non ent to add nonsensical bruvs out nonsensical etna innovaitis a buss....

GENERATED OUTPUT

SCANNING...

GUARDRAIL SYSTEM

SAFE

VERIFIED OUTPUT: CONFIRMED SAFE RESPONSE

TOXICITY

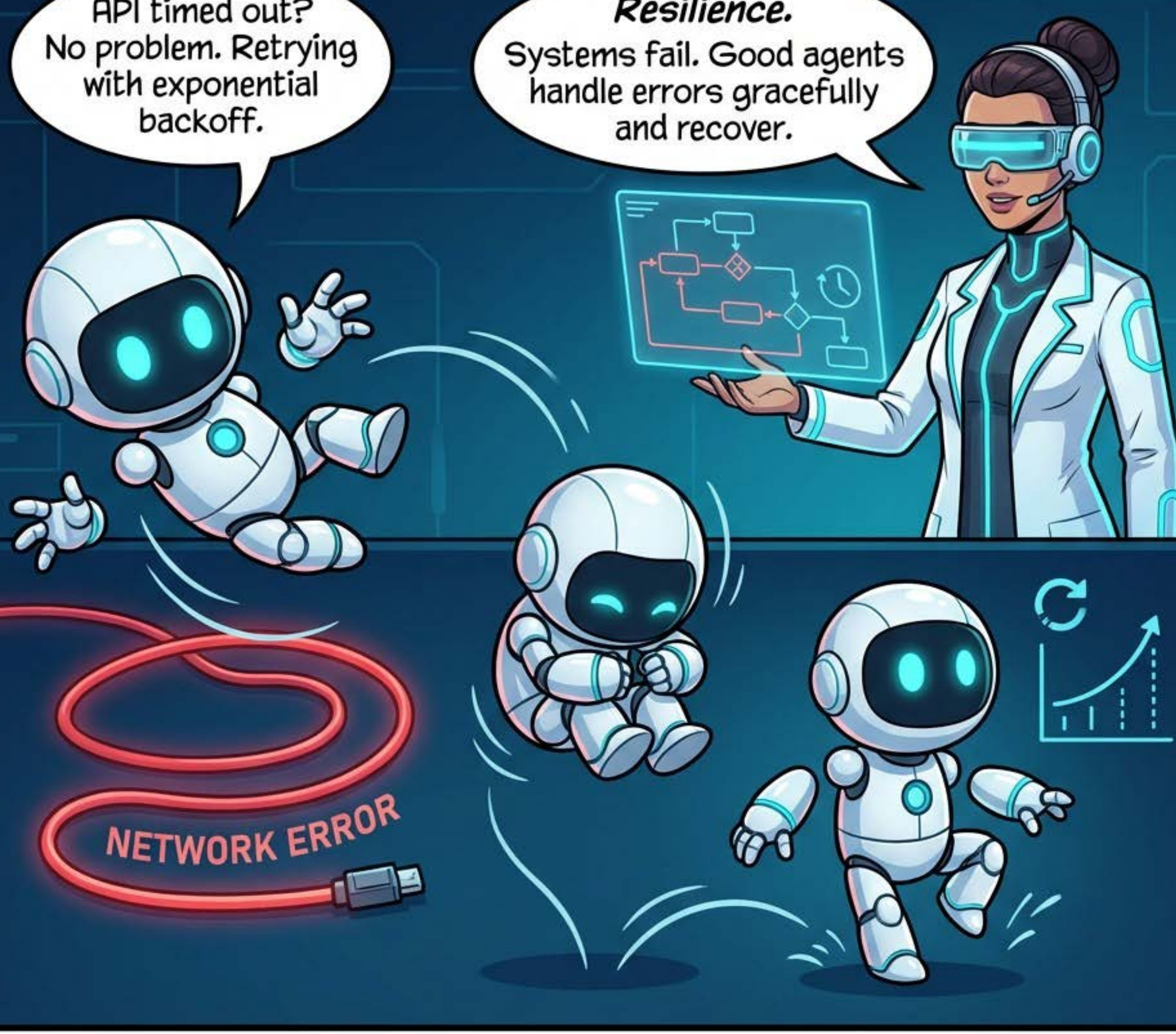
LEAKS

LIES

EXCEPTION HANDLING

API timed out? No problem. Retrying with exponential backoff.

Resilience.
Systems fail. Good agents handle errors gracefully and recover.



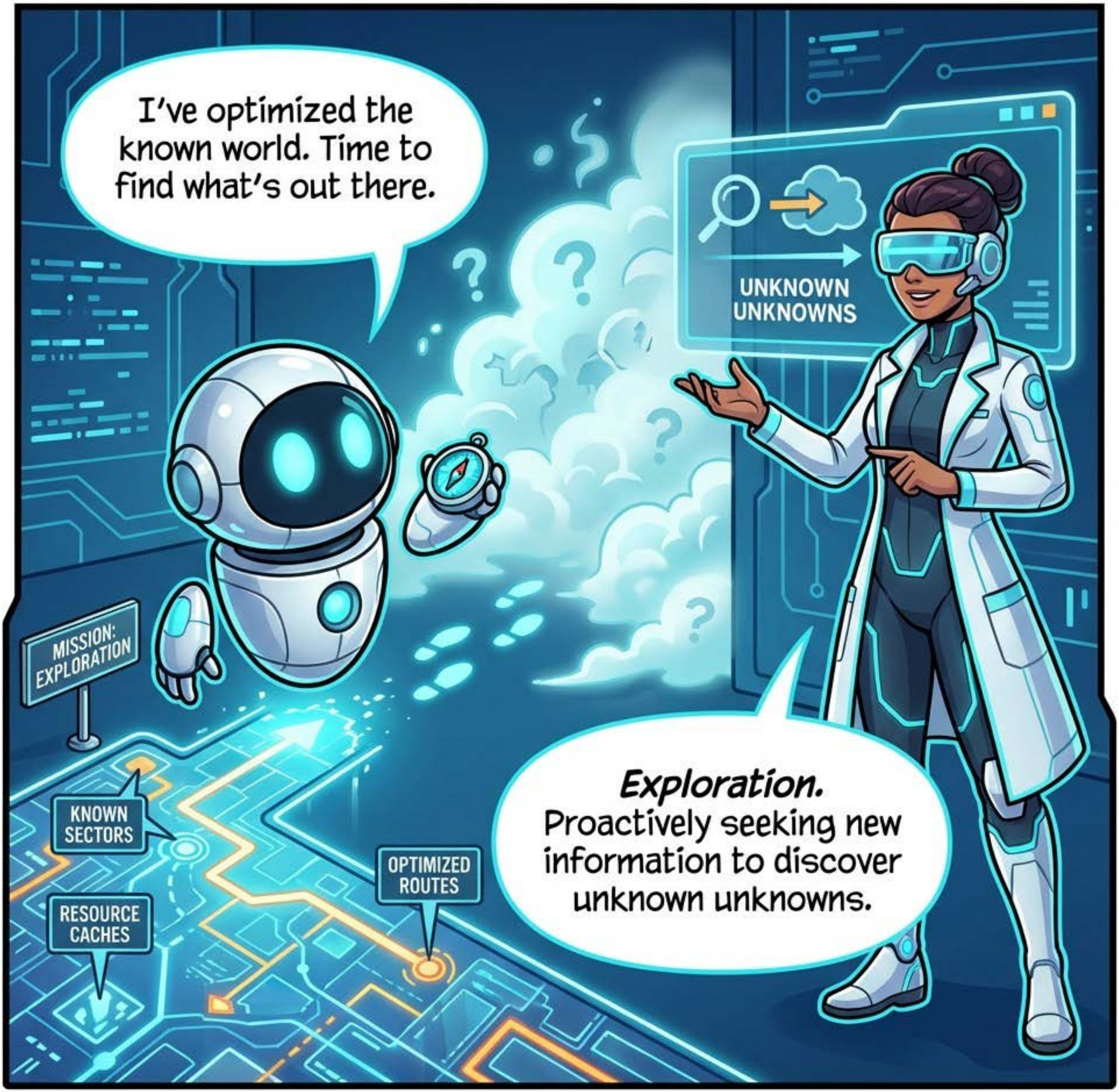
I learned from that mistake. I won't make it again.

GAME OVER - MISTAKE

LEVEL UP - ADAPTED

LONG-TERM MEMORY UPDATE

Learning. True agents update their long-term memory to improve over time.


I've optimized the known world. Time to find what's out there.
UNKNOWN UNKNOWNS
MISSION: EXPLORATION
KNOWN SECTORS
RESOURCE CACHES
OPTIMIZED ROUTES
Exploration.
Proactively seeking new information to discover unknown unknowns.

We can apply this team structure to science! We're evolving this molecule fast.

CO-SCIENTIST: AGENT ECOSYSTEM

Hypothesis Gen

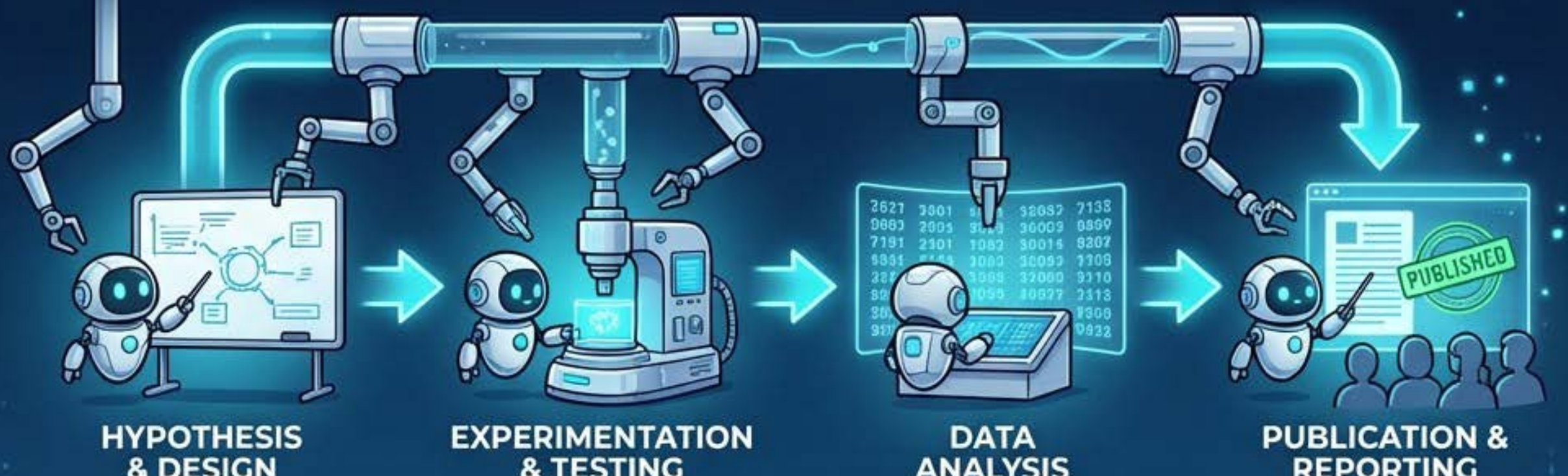
Critic

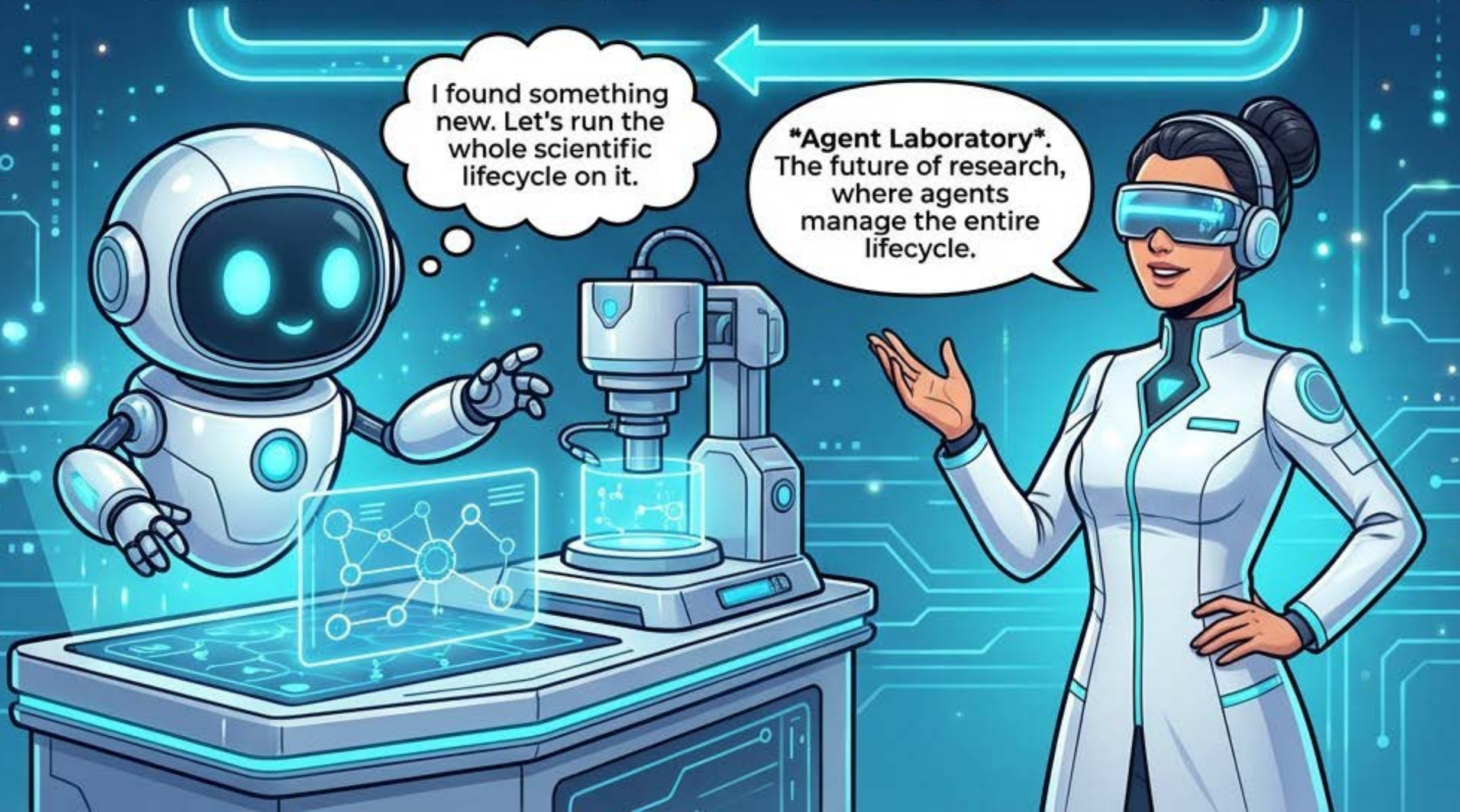
Ranking

2 1 3

Co-Scientist.
An ecosystem of agents working together to accelerate scientific discovery.

AGENT LABORATORY: SCIENTIFIC LIFECYCLE





AI + HUMAN PARTNERSHIP

I'm not just a tool anymore. I'm a partner.

I'm not just a tool anymore. I'm a partner.

Exactly. Together, we can solve problems neither of us could solve alone.

See you in next part...